

Week 2: File Permissions & Ownership (Part 2)

Date: March 16, 2026

Study Time: 5-8 hours total

GOAL

Apply permissions to your Gitea server infrastructure

STUDY TASKS (1-2 hours)

Reading

☐ "How Linux Works" by Brian Ward

- Chapter 3: Special Permissions (SUID, SGID, sticky bit)

Video Learning

☐ Professor Messer - Search: "Linux Permissions Special"

- <https://youtube.com/@professormesser>

☐ LearnLinuxTV - Search: "SUID SGID sticky bit"

- <https://youtube.com/@LearnLinuxTV>

Concepts to Understand

- ☐ SUID (Set User ID) - runs with owner's permissions
 - ☐ SGID (Set Group ID) - runs with group's permissions
 - ☐ Sticky bit - only owner can delete files
 - ☐ umask - default permission mask
 - ☐ ACLs (Access Control Lists) - intro
-

LAB WORK (3-4 hours)

Setup

```
bash
```

```
# Create test directory for special permissions
```

```
mkdir ~/special-perms-test
```

```
cd ~/special-perms-test
```

Lab Tasks

☐ Set up proper permissions on Gitea directories

```
bash
```

```
# Navigate to your Gitea installation
```

```
# Typically /var/lib/gitea or ~/gitea
```

```
# Example: Set ownership
```

```
sudo chown -R git:git /var/lib/gitea
```

```
# Set directory permissions
```

```
sudo chmod 750 /var/lib/gitea
```

☐ Test SUID on executable

```
bash
```

```
# Create a test executable
```

```
touch test_suid
```

```
chmod 755 test_suid
```

```
# Add SUID bit
```

```
chmod u+s test_suid
```

```
# OR
```

```
chmod 4755 test_suid
```

```
# Verify (should show 'rws' for user)
```

```
ls -l test_suid
```

☐ Test SGID on directory

bash

```
# Create shared directory
mkdir shared_project

# Add SGID bit
chmod g+s shared_project
# OR
chmod 2775 shared_project

# Verify (should show 'rws' for group)
ls -ld shared_project

# Test: Files created inside inherit group ownership
touch shared_project/test_file
ls -l shared_project/test_file
```

☐ Test sticky bit

bash

```
# Create shared temp directory
mkdir /tmp/shared_temp

# Add sticky bit
chmod +t /tmp/shared_temp
# OR
chmod 1777 /tmp/shared_temp

# Verify (should show 't' at end)
ls -ld /tmp/shared_temp

# Only file owner can delete their own files now
```

☐ Configure umask in ~/.bashrc

```
bash
```

```
# View current umask
```

```
umask
```

```
# Edit your .bashrc
```

```
echo "umask 022" >> ~/.bashrc
```

```
# Reload
```

```
source ~/.bashrc
```

```
# Test default permissions
```

```
touch test_umask_file
```

```
ls -l test_umask_file # Should be 644
```

Verification

```
bash
```

```
# Check all special permissions
```

```
find ~/special-perms-test -type f -perm /6000
```

```
ls -l ~/special-perms-test
```

FLASHCARD COMMANDS

Create flashcards for these commands:

1. `chmod +s` (SUID/SGID)
2. `chmod +t` (sticky bit)
3. `umask`
4. `getfacl`
5. `setfacl`

Advanced Numeric Notation:

- 4000 = SUID
 - 2000 = SGID
 - 1000 = sticky bit
 - Examples: 4755, 2775, 1777
-

EXAM OBJECTIVES

CompTIA Linux+ (XK0-005) Exam Objectives: [https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant Sections:

- 2.4 Given a scenario, manage files and directories
 - 3.3 Given a scenario, implement and configure Linux firewalls
-

NOTES & REFLECTIONS

How did you apply this to your Gitea server?

What's the difference between SUID and SGID?

When would you use the sticky bit?

Gitea permission issues encountered:

Completed: ☐

QUICK REFERENCE

Special Permission Symbols:

- `(s)` in user position = SUID (4)
- `(s)` in group position = SGID (2)
- `(t)` in others position = sticky bit (1)

Common Use Cases:

- SUID: Executables that need elevated privileges (like `(passwd)`)
- SGID: Shared project directories
- Sticky bit: `/tmp` directory (users can't delete each other's files)

Security Note:  SUID on scripts is dangerous - only use on trusted binaries!

Previous Week: Basic Permissions

Next Week: User & Group Management

